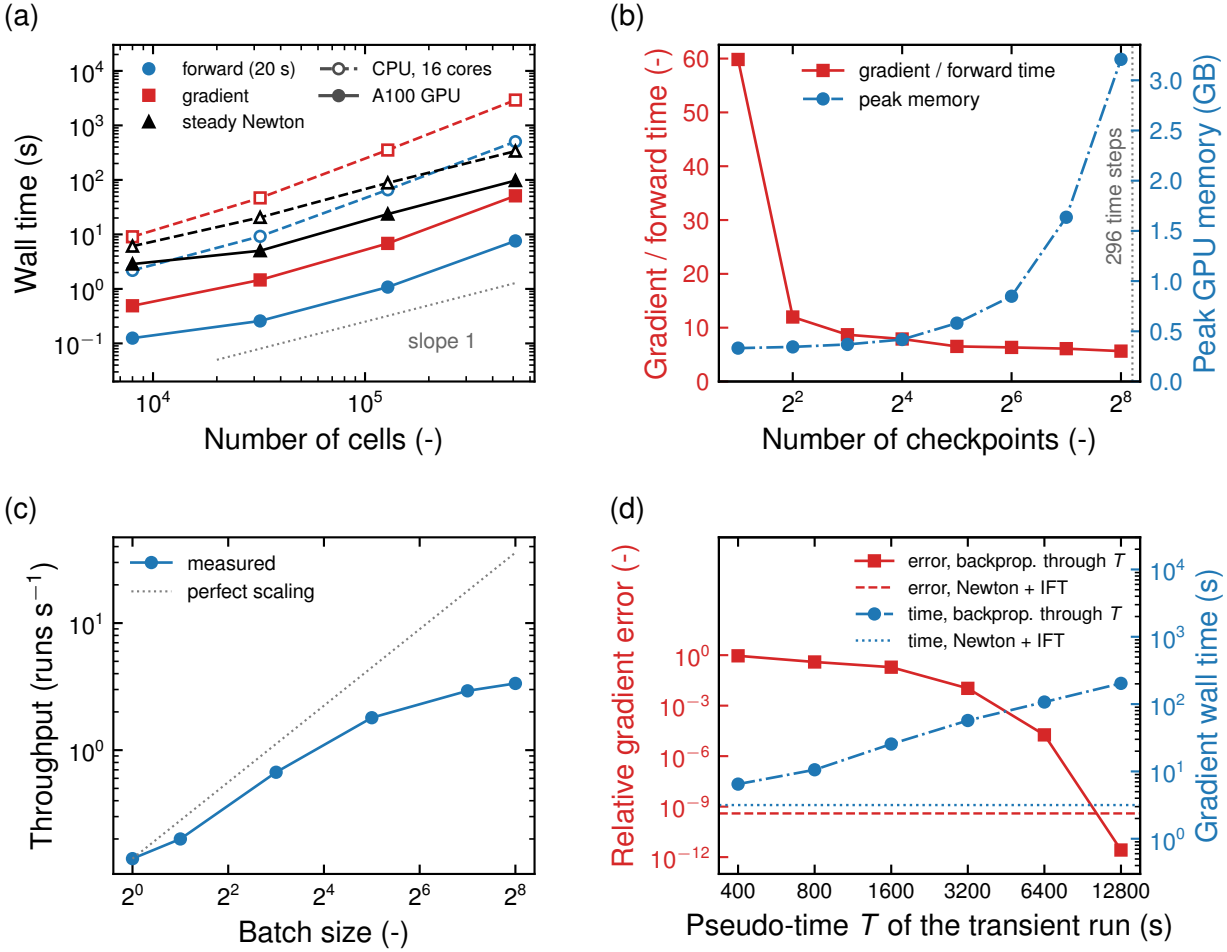
(a)
forward (20 s)
gradient
steady Newton
CPU, 16 cores
A100 GPU
slope 1
Wall time (s)
Number of cells (-)
(b)
gradient / forward time
peak memory
296 time steps
Gradient / forward time (-)
Peak GPU memory (GB)
Number of checkpoints (-)
(c)
measured
perfect scaling
Throughput (runs s^{-1})
Batch size (-)
(d)
error, backprop. through T
error, Newton + IFT
time, backprop. through T
time, Newton + IFT
Relative gradient error (-)
Gradient wall time (s)
Pseudo-time T of the transient run (s)